Slide 1 – Introduction

"Good evening everyone.

I'm **V. Prashoharan**, and today our team is presenting our E-commerce project proposal titled **PawMart – A One-Stop Web Marketplace for Pet Products and Pet Care Services.**

Our team consists of five members:
V. Prashoharan,
Bashini Rashmika Vindyani,
J.P. Inuka Sithanja Jayasinghe,
D.S.N. Vitharana,
and B.R.G.S. Sandaruwan.

The main goal of this project is to provide pet owners with one centralized platform where they can purchase pet products, manage their pet's health information, and book pet care services without using multiple different applications.

Now I'd like to explain the problem we identified and how our solution addresses it."

Slide 2 – Market Gap & Proposed Solution (≈1 minute)

"Before deciding on this project, we looked at how pet owners currently manage their pets.

We found that most people use separate platforms for different tasks. For example, they might buy food from one website, contact a veterinary clinic through social media, and manually keep track of vaccination dates.

This makes the overall process inconvenient and sometimes causes owners to forget important appointments or vaccinations.

To solve this problem, we propose PawMart.

PawMart combines online shopping, appointment booking, pet health records, and a smart pet care assistant into one responsive web application.

Instead of switching between multiple services, users can manage everything from a single platform."

Slide 3 – Project Objectives & Outcomes (≈45 seconds)

"Our project has three main objectives.

The first is to create a centralized marketplace that combines pet products and pet care services.

The second is to automate repetitive tasks such as vaccination reminders by sending email notifications.

The third is to improve the user experience by providing a Smart Pet Care Assistant that gives general guidance based on predefined rules.

By the end of this project, we expect to deliver a fully functional MVP that is deployed using free cloud platforms and completed within our eight-week development schedule."

Slide 4 – Core Platform Features (≈1 minute)

"The core functionality of PawMart is divided into three main areas.

First is secure authentication. Users can register and log in using JWT authentication, while passwords are securely stored using bcrypt hashing.

Second is the e-commerce module, where users can browse products, add items to their cart, place orders, and leave reviews after purchasing.

Finally, we have separate dashboards for different user roles.

Admins can manage the entire marketplace, while vendors can add products, update inventory, and monitor their own sales.

These features form the foundation of our application."

Slide 5 – Innovative Functional Modules (≈1 minute)

"Besides the standard e-commerce features, we wanted to include a few practical features that add value while remaining achievable within our project timeline.

The first is the Smart Pet Care Assistant.

Instead of using expensive AI services, our assistant is rule-based. It provides breed-specific care advice, feeding recommendations, and suggests suitable products based on the user's inputs.

The second feature is Pet Health Records and Vaccination Reminders.

Each pet has its own digital profile where information such as vaccinations, allergies, age, and weight can be stored.

Based on these records, the system automatically sends reminder emails before vaccination dates.

The third feature is Appointment Booking, which allows pet owners to schedule appointments with veterinarians or grooming service providers through the platform."

Slide 6 – Strategic Technology Choices (≈1 minute)

"For the technology stack, we intentionally selected tools that are free, reliable, and widely used in industry.

For the frontend, we selected React.js with Tailwind CSS to build a modern and responsive interface.

For the backend, we use Node.js with Express.js because they are lightweight and work efficiently with REST APIs.

MongoDB Atlas is our database because it provides a flexible document structure, which is useful for handling different types of pet information and products.

Cloudinary is used for image storage, while Nodemailer handles email notifications such as vaccination reminders.

For deployment, we selected Vercel for the frontend and Render for the backend.

Since this is an academic project, we use a simulated payment module instead of integrating a real payment gateway."

Slide 7 – Use Case Diagram (≈45 seconds)

"This diagram shows how different users interact with the system.

The main actor is the Pet Owner, who can register, browse products, manage pet profiles, place orders, book appointments, and receive vaccination reminders.

Vendors manage products and inventory.

Service providers manage appointments.

Administrators manage users, vendors, and system reports.

Overall, this diagram summarizes the major functionalities available for each type of user."

Slide 8 – ER Diagram (≈45 seconds)

"This ER diagram represents our database design.

The central entity is the User.

Each user can own multiple pets, place orders, book appointments, and submit reviews.

Each pet has a corresponding health record that stores medical information.

Orders are connected to products through order items, while appointments link users, pets, and service providers together.

This database structure ensures that all system modules remain properly connected."

Slide 9 – System Architecture (≈45 seconds)

"This diagram illustrates the overall architecture of PawMart.

Users interact with the React web application.

All requests are sent to the Express REST API.

JWT authentication validates user access before requests reach the business logic layer.

The business logic communicates with MongoDB Atlas to store and retrieve data.

It also connects to Cloudinary for image management and Nodemailer for sending reminder emails.

This layered architecture makes the system modular and easier to maintain."

Slide 10 – Activity Diagram (≈45 seconds)

"This activity diagram demonstrates the customer purchasing process.

The user first logs in and browses products.

If needed, they can use the Smart Pet Care Assistant for recommendations.

They then add products to the shopping cart and proceed to checkout.

The payment process is simulated.

If payment succeeds, the order is created, a confirmation email is sent, and the order status is updated.

If payment fails, the user simply retries the payment process."

Slide 11 – Database Design & Justification (≈50 seconds)

"We selected MongoDB Atlas because it fits well with the MERN stack and provides a flexible document-based structure.

Our main collections include Users, Pets, Products, Orders, Payments, Reviews, Services, Appointments, and Pet Health Records.

MongoDB allows us to manage different types of data without requiring complex schema changes, which makes development faster and more suitable for our project."

Slide 12 – Team Responsibilities (≈40 seconds)

"Our work has been divided equally among all five team members.

I will be responsible for authentication, authorization, and the admin dashboard.

Bashini will develop product management and the vendor dashboard.

Inuka will implement the shopping cart, checkout process, and simulated payment module.

Vitharana will develop order management and appointment booking.

Finally, Sandaruwan will work on reviews, the Smart Pet Care Assistant, and pet health records with vaccination reminders.

This distribution allows every member to contribute equally."

Slide 13 – 8-Week Development Timeline (≈40 seconds)

"Since all of us are currently doing internships, we planned a realistic eight-week schedule.

The first two weeks focus on planning, UI design, authentication, and database setup.

The middle weeks cover product management, shopping cart, payments, appointment booking, and health records.

Finally, the last two weeks are allocated for integrating all modules, testing, fixing issues, deployment, and preparing for the final demonstration.

This schedule ensures that the project remains achievable within the given timeframe."

Slide 14 – Closing (≈20 seconds)

"To conclude, PawMart is a practical and realistic web application that combines e-commerce and pet care services into a single platform.

We believe this project addresses a genuine problem faced by pet owners while remaining achievable within our academic timeline.

Thank you very much for your time and attention.

We are happy to answer any questions."